Project Report: Fine-Tuning a Medical Language Model

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Tools Used: Python, yFinance, pandas, matplotlib, seaborn, statsmodels, scikit-learn

Project Title

Medical Language Model Fine-Tuning using Unsloth and HuggingFace

Objective

To fine-tune a pre-trained LLaMA-3 8B model on a medical dataset to enhance its performance in clinical reasoning and chain-of-thought (CoT) explanations. This can support use-cases in medical assistant tools, automated diagnostics, and medical Q&A systems.

Tools & Libraries Used

- **Unsloth**: For optimized fine-tuning of LLaMA models.
- **Transformers (HuggingFace)**: Model and tokenizer utilities.
- **Datasets (HuggingFace)**: To load and preprocess the training dataset.
- **WandB**: Experiment tracking.
- **PEFT (Parameter Efficient Fine-Tuning)**: To use LoRA.
- **bitsandbytes**: 8-bit quantization for memory efficiency.

Model Setup

Model Chosen

- **LLaMA-3 8B (8-bit)** from `unsloth`

Fine-Tuning Strategy

- **LoRA**: Used to apply lightweight adapters during training.
- Target Modules: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`

Dataset

Source

- Dataset loaded from HuggingFace Hub: `khanrc/medical_meadow_chain_of_thought`

Features

- **`question`**: Medical question.
- **`answer`**: Final answer.
- **`cot`**: Chain of Thought explanation (used for instruct fine-tuning).

Training Configuration

- **Epochs**: 3
- **Batch Size**: 2
- **Sequence Length**: 2048 tokens
- **Learning Rate**: 2e-4
- **Logging**: Using WandB
- **Evaluation**: Performed every 300 steps
- **Save Strategy**: Saves every 500 steps
- **LoRA Rank**: 16

Monitoring

- **WandB** dashboard used for tracking loss, learning curves, and memory usage.
- Model uses `bf16` precision for faster and memory-efficient training.

Results & Evaluation

While the notebook is primarily focused on the setup and training loop, the results section could include:

- Loss convergence graphs (tracked on WandB).
- Sample generations (qualitative evaluation).
- Rouge or BLEU scores if provided in evaluation scripts.

Observations & Future Work

Strengths

- Efficient fine-tuning using LoRA and 8-bit quantization.
- Real-world medical data with Chain of Thought enhances reasoning capabilities.
- Lightweight enough to run on consumer-grade GPUs.

Improvements

- Add evaluation on a test set to quantify accuracy.
- Include instruction tuning with real patient dialogues.
- Add safety filtering for medical hallucinations.

Links

- **HuggingFace Dataset**: `khanrc/medical_meadow_chain_of_thought`
- **Tracking Dashboard**: Weights & Biases project